

Luiz M. Faria

Ph.D. in Applied Mathematics

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Research Interests

- Numerical methods for PDEs, asymptotic analysis, fluid dynamics, combustion theory, partial differential equations, dynamical systems, stability theory.

Education

- **King Abdullah University of Science and Technology (KAUST)** Thuwal, KSA
Ph.D. in Applied Math. & Computational Science (GPA: 4.0/4.0) 01/2012—12/2014
– Thesis: Qualitative and asymptotic theory of detonations.
– Advisor: Aslan Kasimov.
- **King Abdullah University of Science and Technology (KAUST)** Thuwal, KSA
M.Sc. in Applied Math. & Computational Science (GPA: 3.89/4.0) 08/2010—12/2011
- **Texas A&M University** College Station, Texas, USA
B.Sc. in Mathematics with Honors (GPA: 4.0/4.0) 08/2006—05/2010

Professional experience

- **INRIA Research Scientist** Ile-de-France, France
10/2020—present
- **INRIA Starting Research Position** Ile-de-France, France
01/2019—09/2020
- **Ingénieur d'étude métier at CS Communication & Systèmes** Ile-de-France, France
07/2018—12/2018
- **Instructor in Applied Math at MIT** Cambridge, MA, USA
06/2015—06/2018

Supervision of research activities

- **Student's name:** Alexis Anne. **Starting date:** 06/2021. **End date:** 08/2019. **Description:** Supervisor for second-year student in a “stage de PRE” context at ENSTA. Project involved studying different truncation methods for free-surface gravity waves with a particular focus on techniques which work for boundary integral equations. Outcome was a written report.
- **Student's name:** Chedhli Masmoudi. **Starting date:** 05/2019. **End date:** 07/2019. **Description:** Co-supervisor for second-year student in a “stage de PRE” context at ENSTA. Project involved studying the diffraction by unbounded interfaces in a boundary integral formulation. In particular, different truncation techniques were employed, with special focus on analytic dilation techniques similar to the method of Perfectly-Matched-Layers (PMLs). Outcome was a written report.
- **Student's name:** Shem Lau-Chapdelaine. **Starting date:** 11/2016. **End date:** 03/2017. **Description:** Main supervisor for exchange visiting student. Project involved using simplified models to study a wave-propagation problem in combustion theory. Outcome was a peer-reviewed conference paper.
- **Student's name:** Max Vargas. **Starting date:** 05/2016. **End date:** 08/2016. **Description:** Supervisor for MIT undergraduate research program (UROP). Project involved finding a special type of traveling wave solutions for a simplified combustion model. Outcome was an internal project report.

Computer and language skills

Languages: Native in Portuguese, fluent in Spanish, English and French.

Programming: C++, Julia, Python, Fortran, Matlab.

Teaching

- Numerical linear algebra, ENSTA 2021.
- Finite elements and boundary elements, ENSTA 2020.
- Scientific computing in C++, ENSTA 2020.
- Finite Element Method, ENSTA 2019.
- Principles of Continuum Applied Mathematics; MIT, Spring 2016, Spring 2017, and Spring 2018.
- Project Lab in Mathematics; MIT, Fall 2017.
- Seminar in Physical Applied Mathematics; MIT, Fall 2016.
- Recitation leader for Ordinary Differential Equations; MIT, Fall 2015.
- Teaching assistant for Applied PDE I; KAUST, Fall 2011.
- Teaching assistant for Single Variable/Multivariable Calculus; Texas A&M, Fall 2007/Spring 2010.
- Mentor for Advanced Calculus; Texas A&M, Spring 2010.

Publications

1. L. M. Faria, C. Pérez-Arancibia, and M. Bonnet. General-purpose kernel regularization of boundary integral equations via density interpolation. *Computer Methods in Applied Mechanics and Engineering*, 378:113703, 2021
2. J. Melguizo-Gavilanes, Y. Balossier, and L. M. Faria. Experimental and theoretical observations on ddt in smooth narrow channels. *Proceedings of the Combustion Institute*, 38(3):3497–3503, 2021
3. C. Pérez-Arancibia, C. Turc, L. M. Faria, and C. Sideris. Planewave density interpolation methods for the efie on simple and composite surfaces. *IEEE Transactions on Antennas and Propagation*, 2020
4. I. Labarca, L. M. Faria, and C. Pérez-Arancibia. Convolution quadrature methods for time-domain scattering from unbounded penetrable interfaces. *Proceedings of the Royal Society A*, 475(2227):20190029, 2019
5. C. Pérez-Arancibia, C. Turc, and L. Faria. Planewave density interpolation methods for 3d helmholtz boundary integral equations. *SIAM Journal on Scientific Computing*, 41(4):A2088–A2116, 2019
6. C. Pérez-Arancibia, L. M. Faria, and C. Turc. Harmonic density interpolation methods for high-order evaluation of laplace layer potentials in 2d and 3d. *Journal of Computational Physics*, 376:411–434, 2019
7. D. M. Harris, P.-T. Brun, A. Damiano, L. M. Faria, and J. W. Bush. The interaction of a walking droplet and a submerged pillar: From scattering to the logarithmic spiral. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 28(9):096105, 2018
8. O. Mickelin, J. Słomka, K. J. Burns, D. Lecoanet, G. M. Vasil, L. M. Faria, and J. Dunkel. Anomalous chained turbulence in actively driven flows on spheres. *Physical review letters*, 120(16):164503, 2018

9. G. Pucci, D. M. Harris, L. M. Faria, and J. W. Bush. Walking droplets interacting with single and double slits. *Journal of Fluid Mechanics*, 835:1136–1156, 2018.
10. L. M. Faria. A model for Faraday pilot waves over variable topography. *Journal of Fluid Mechanics*, 811:51–66, 2017.
11. L. M. Faria and R. R. Rosales. Equation level matching: An extension of the method of matched asymptotic expansion for problems of wave propagation. *Studies in Applied Mathematics*, 139(2):265–287, 2017.
12. G. Pucci, G. J. Sáenz, L. M. Faria, and J. W. M. Bush. Non-specular reflection of walking droplets. *Journal of Fluid Mechanics*, 804:R3, 2016.
13. L. M. Faria, A. R. Kasimov, and R. R. Rosales. Study of a model equation in detonation theory: Multidimensional effects. *SIAM Journal on Applied Mathematics*, 76(3):887–909, 2016.
14. R. Semenko, L. M. Faria, A. R. Kasimov, and B. S. Ermolaev. Set-valued solutions for non-ideal detonation. *Shock Waves*, 26(2):141–160, 2016.
15. L. M. Faria, A. R. Kasimov, and R. R. Rosales. Theory of weakly nonlinear self-sustained detonations. *Journal of Fluid Mechanics*, 784:163–198, 2015.
16. L. M. Faria and A. R. Kasimov. Qualitative modeling of the dynamics of detonations with losses. *Proceedings of the Combustion Institute*, 35:2015–2023, 2015.
17. L. M. Faria, A. R. Kasimov, and R. R. Rosales. Study of a Model Equation in Detonation Theory. *SIAM Journal on Applied Mathematics*, 74(2):547–570, 2014.
18. A. R. Kasimov, L. M. Faria, and R. R. Rosales. Model for Shock Wave Chaos. *Physical Review Letters*, 110(10):104104, 2013.

Conferences and Invited talks

1. “La simulation numérique au service des phénomènes physiques”. Unithé ou café, INRIA Saclay, 2021.
2. “Boundary integral equation methods on dynamic surfaces”. INRIA-Skoltech Workshop, 2020.
3. “Mathematical modeling and numerical simulations of Faraday pilot waves”. PUC, 2020.
4. “Computing the Mandelbrot set in Julia”. SED coding dojo, INRIA Saclay, 2019.
5. “Harmonic density interpolation methods for high-order evaluation of Laplace layer potentials in 2D and 3D”. ENSTA, 2019.
6. “A model for Faraday pilot-waves over variable topography”. Universidad Autonoma de Mexico, May 2017.
7. “Asymptotic theory of weak heat release combustion”. New Jersey Institute of Technology, April 2017.
8. “Weakly nonlinear combustion waves”. Rensselaer Polytechnic Institute, April 2016.
9. “Qualitative and asymptotic theory of detonations”. IMPA, Brazil, 2015.
10. “Qualitative modeling of the dynamics of detonation with losses”. 35th International Symposium on Combustion. San Francisco, California, August 2014
11. “Qualitative and asymptotic theory of detonations”. SIAM Nonlinear Waves and Coherent Structures. Cambridge, UK, August 2014

12. “An asymptotic theory of weakly non-linear detonations”. KAUST selected student representative at SIAM Annual Meeting, Chicago, Illinois, July 2014
13. “A nonlocal hyperbolic PDE for chaotic shocks”. SIAM Conference on Analysis of Partial Differential Equations, Lake Buena Vista, Florida, December 2013
14. “A simple model for spherical detonation”. 24th International Colloquium on Dynamics of Explosions and Reactive Systems. Taipei, Taiwan, July 2013
15. “A model for shock wave chaos” 65th Annual Meeting of the APS Division of Fluid Dynamics. San Diego, California, November 2012
16. ”A model for shock wave chaos” 14th International Conference on Hyperbolic Problems:Theory, Numerics, Applications. University of Padova, Italy, June 2012

Community responsibilities

- Member of the organizing committee for the next WAVES conference to take place at POEMS in 2021.
- Co-organizer of the Numerical Methods for PDE seminar at MIT (<http://math.mit.edu/seminars/nmpde/>) from 08/2015 until 05/2018.

Awards

- Participant of ANR JCJC, project FAsTD (Flame Acceleration and Transition to Detonation in Narrow Channels). Start date: 12/2020. End date: 12/2024. Coordinator: J. Melguizo Gavilanes (Institute Pprime).
- National Science Foundation Award, 2017. Co-PI with J.W.M. Bush and P. Saenz
- KAUST Graduate Fellowship, 2010 – 2014
- Elected SIAM Student Representative from KAUST, 2014
- Dean’s Award for Academic Excellence from KAUST, 2011
- KAUST Discovery Scholarship, 2007 – 2010
- Classroom Excellence Award, Texas A&M, 2008 – 2009
- Gold Medal on National Physics Olympiad, Panama, XI grade, 2004

References

- [1] L. M. Faria. A model for Faraday pilot waves over variable topography. *Journal of Fluid Mechanics*, 811:51–66, 2017.
- [2] L. M. Faria and A. R. Kasimov. Qualitative modeling of the dynamics of detonations with losses. *Proceedings of the Combustion Institute*, 35:2015–2023, 2015.
- [3] L. M. Faria, A. R. Kasimov, and R. R. Rosales. Study of a Model Equation in Detonation Theory. *SIAM Journal on Applied Mathematics*, 74(2):547–570, 2014.
- [4] L. M. Faria, A. R. Kasimov, and R. R. Rosales. Theory of weakly nonlinear self-sustained detonations. *Journal of Fluid Mechanics*, 784:163–198, 2015.
- [5] L. M. Faria, A. R. Kasimov, and R. R. Rosales. Study of a model equation in detonation theory: Multidimensional effects. *SIAM Journal on Applied Mathematics*, 76(3):887–909, 2016.
- [6] L. M. Faria, C. Pérez-Arancibia, and M. Bonnet. General-purpose kernel regularization of boundary integral equations via density interpolation. *Computer Methods in Applied Mechanics and Engineering*, 378:113703, 2021.
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- [13] C. Pérez-Arancibia, L. M. Faria, and C. Turc. Harmonic density interpolation methods for high-order evaluation of laplace layer potentials in 2d and 3d. *Journal of Computational Physics*, 376:411–434, 2019.
- [14] C. Pérez-Arancibia, C. Turc, and L. Faria. Planewave density interpolation methods for 3d helmholtz boundary integral equations. *SIAM Journal on Scientific Computing*, 41(4):A2088–A2116, 2019.
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- [16] G. Pucci, D. M. Harris, L. M. Faria, and J. W. Bush. Walking droplets interacting with single and double slits. *Journal of Fluid Mechanics*, 835:1136–1156, 2018.

- [17] G. Pucci, G. J. Sáenz, L. M. Faria, and J. W. M. Bush. Non-specular reflection of walking droplets. *Journal of Fluid Mechanics*, 804:R3, 2016.
- [18] R. Semenko, L. M. Faria, A. R. Kasimov, and B. S. Ermolaev. Set-valued solutions for non-ideal detonation. *Shock Waves*, 26(2):141–160, 2016.